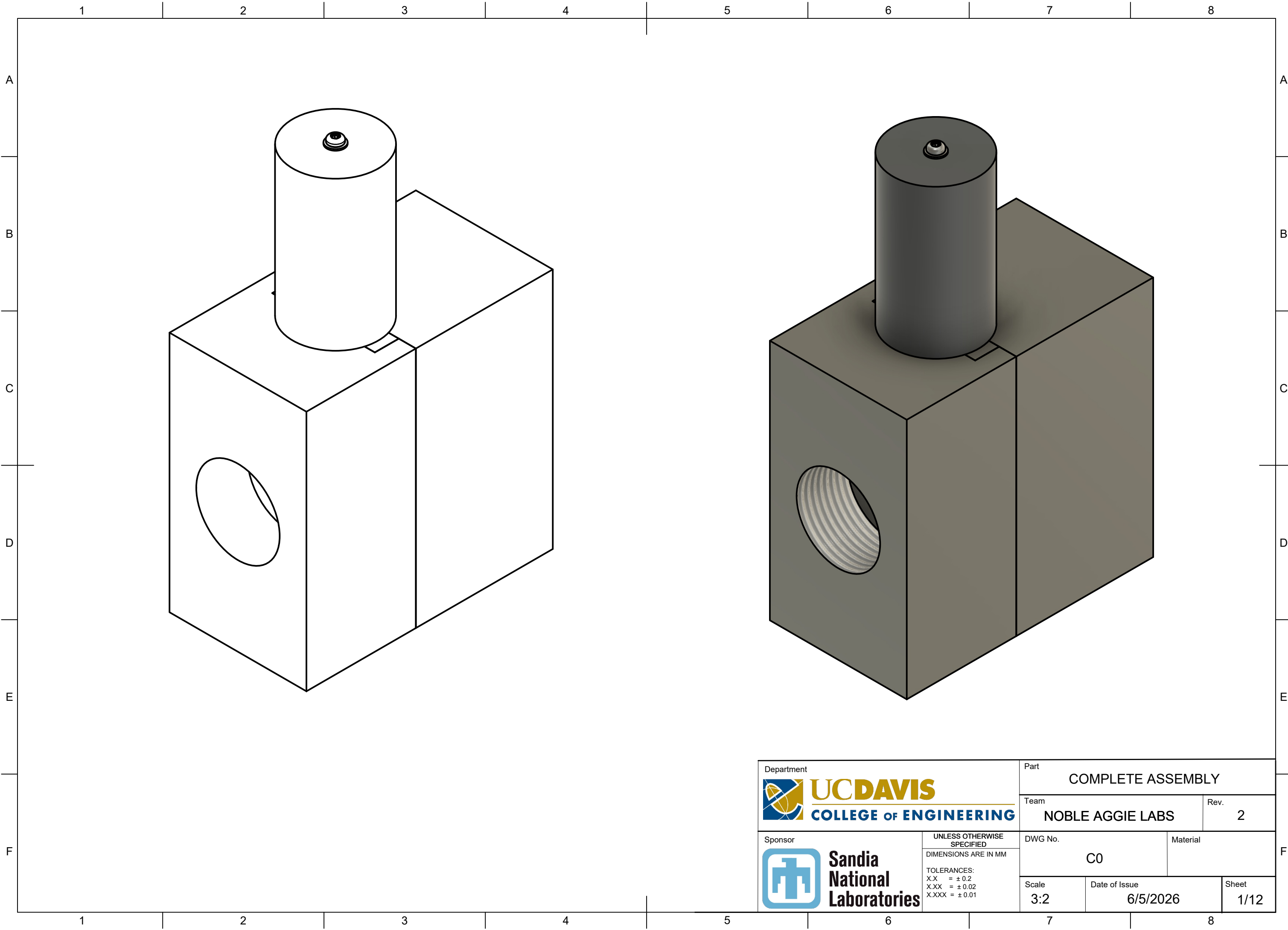
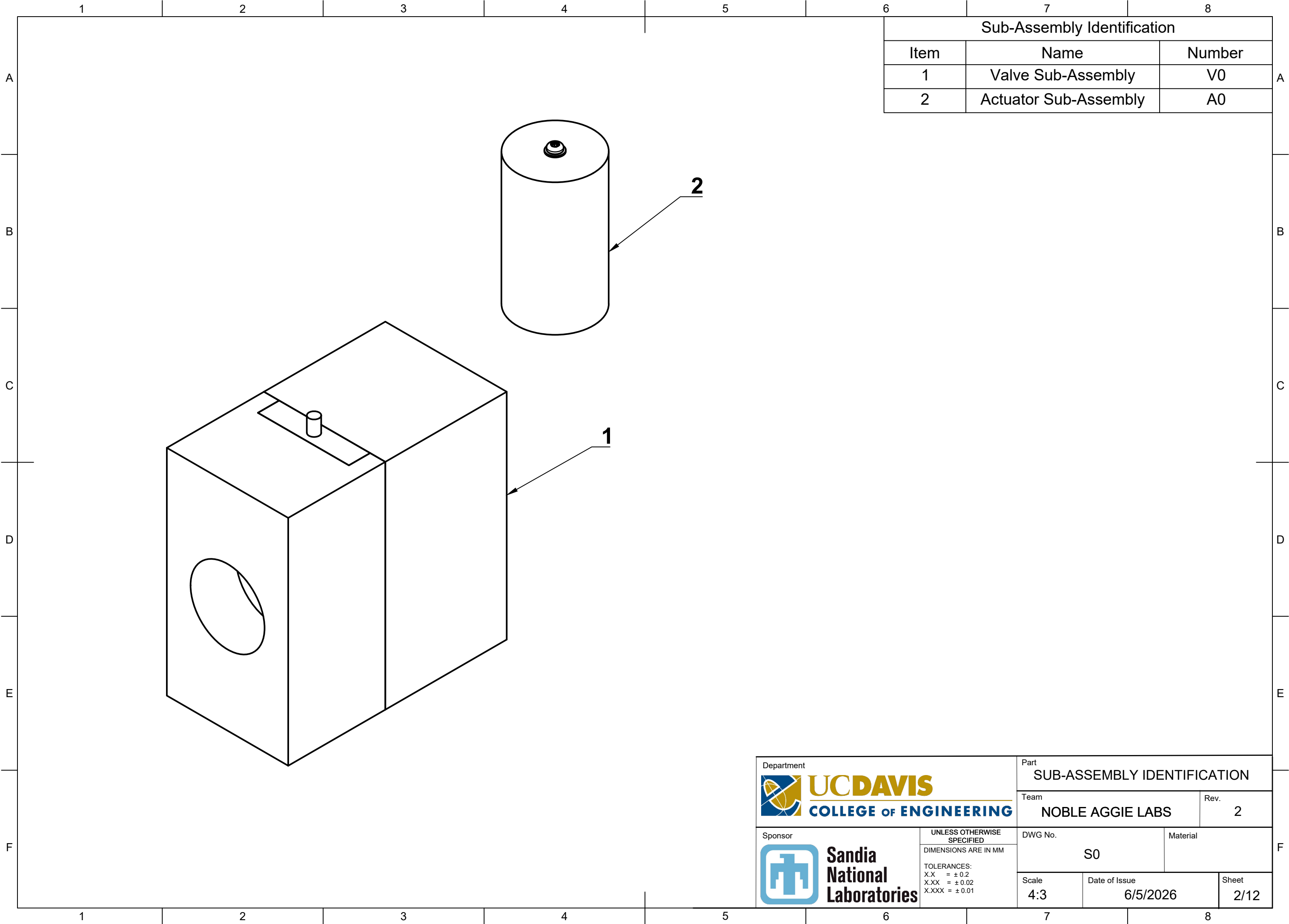


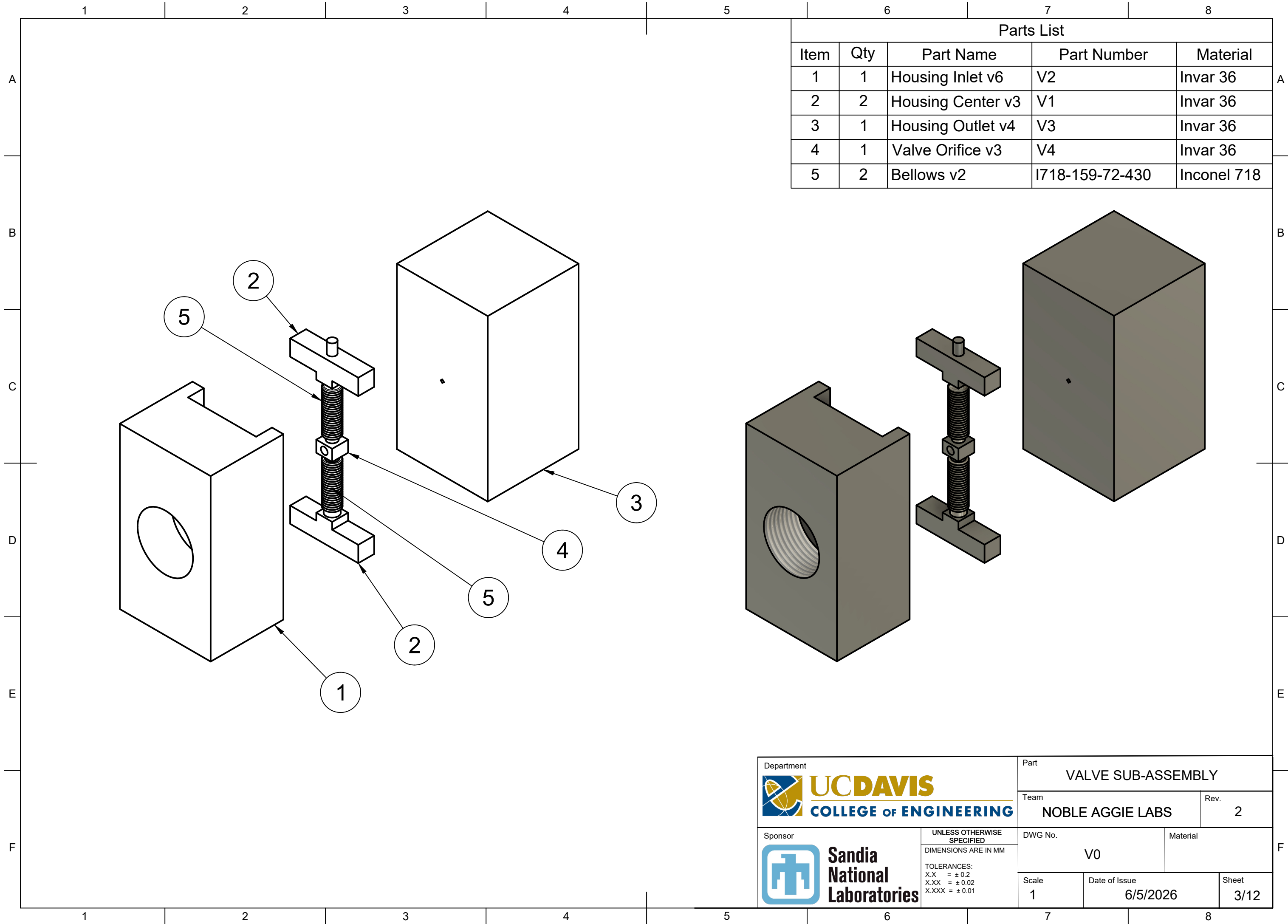


Department		Part	
		COMPLETE ASSEMBLY	
Team		NOBLE AGGIE LABS	Rev. 2
Sponsor	UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM	DWG No.	Material
 Sandia National Laboratories	TOLERANCES: X.X = ± 0.2 X.XX = ± 0.02 X.XXX = ± 0.01	C0	
Scale		Date of Issue	Sheet
3:2		6/5/2026	1/12



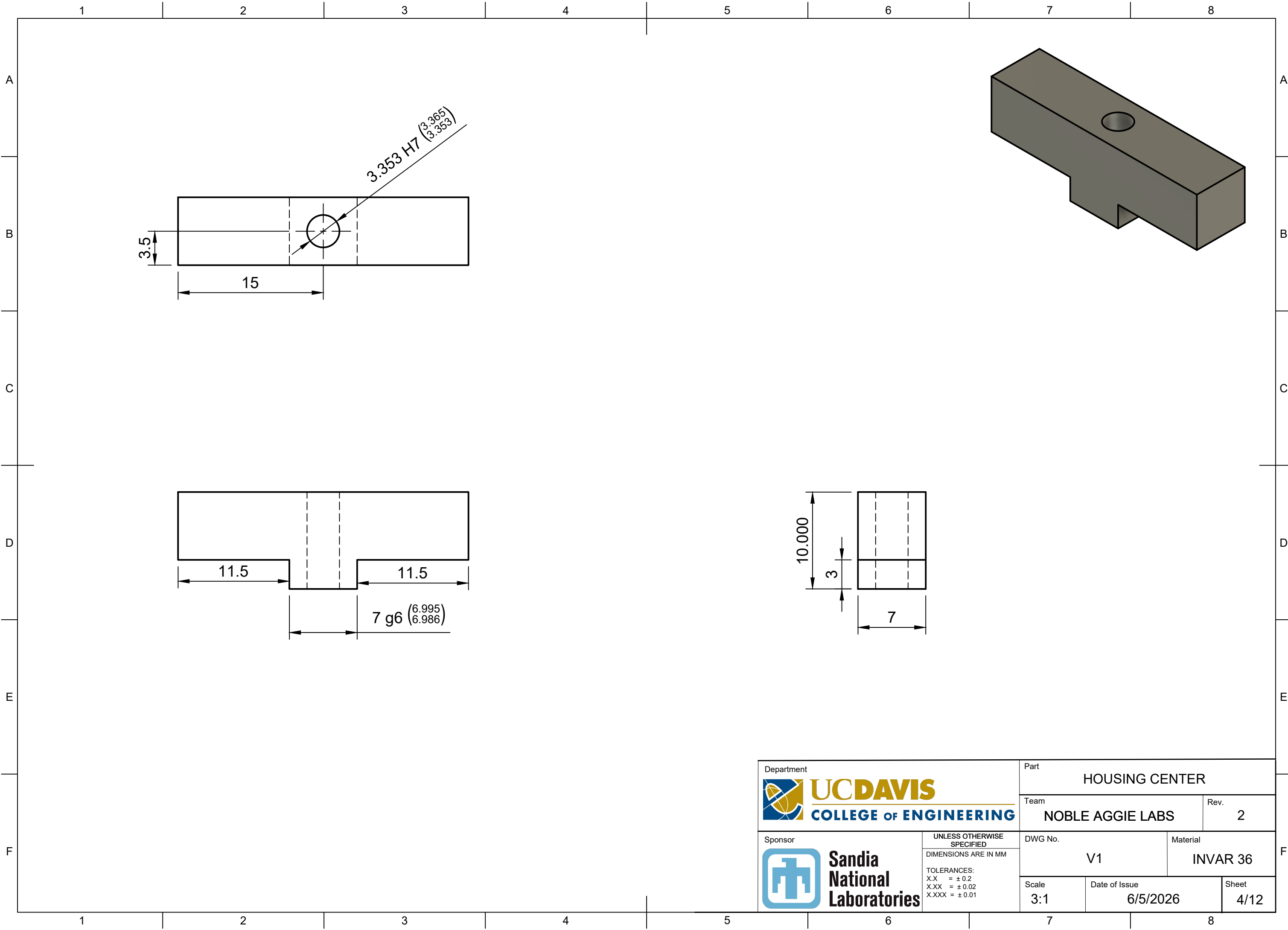
Sub-Assembly Identification		
Item	Name	Number
1	Valve Sub-Assembly	V0
2	Actuator Sub-Assembly	A0

Department		Part	
		SUB-ASSEMBLY IDENTIFICATION	
Sponsor		Team	Rev.
		NOBLE AGGIE LABS	2
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM		DWG No.	Material
TOLERANCES: X.X = ± 0.2 X.XX = ± 0.02 X.XXX = ± 0.01		S0	
Scale		Date of Issue	Sheet
4:3		6/5/2026	2/12

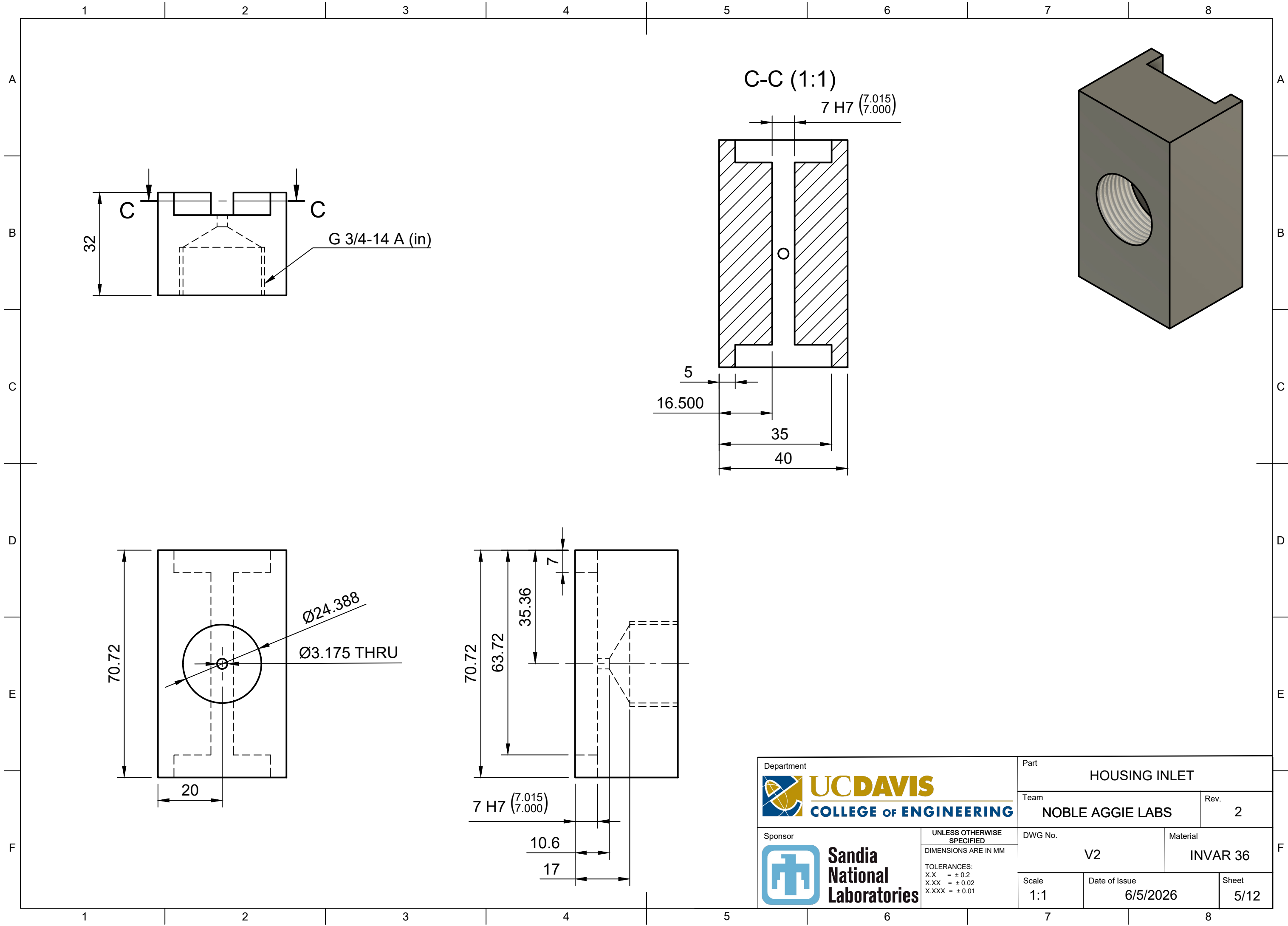


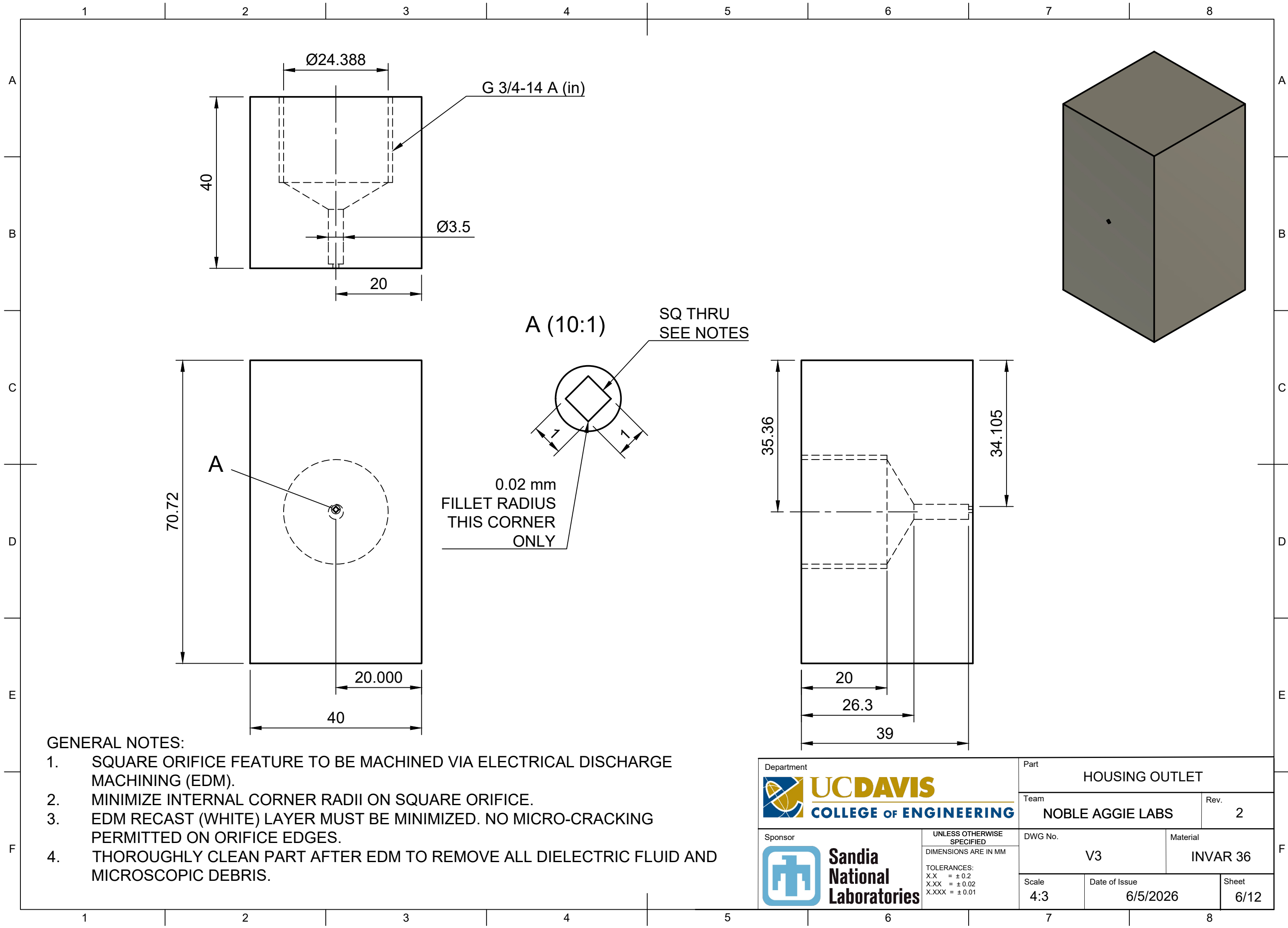
Parts List				
Item	Qty	Part Name	Part Number	Material
1	1	Housing Inlet v6	V2	Invar 36
2	2	Housing Center v3	V1	Invar 36
3	1	Housing Outlet v4	V3	Invar 36
4	1	Valve Orifice v3	V4	Invar 36
5	2	Bellows v2	I718-159-72-430	Inconel 718

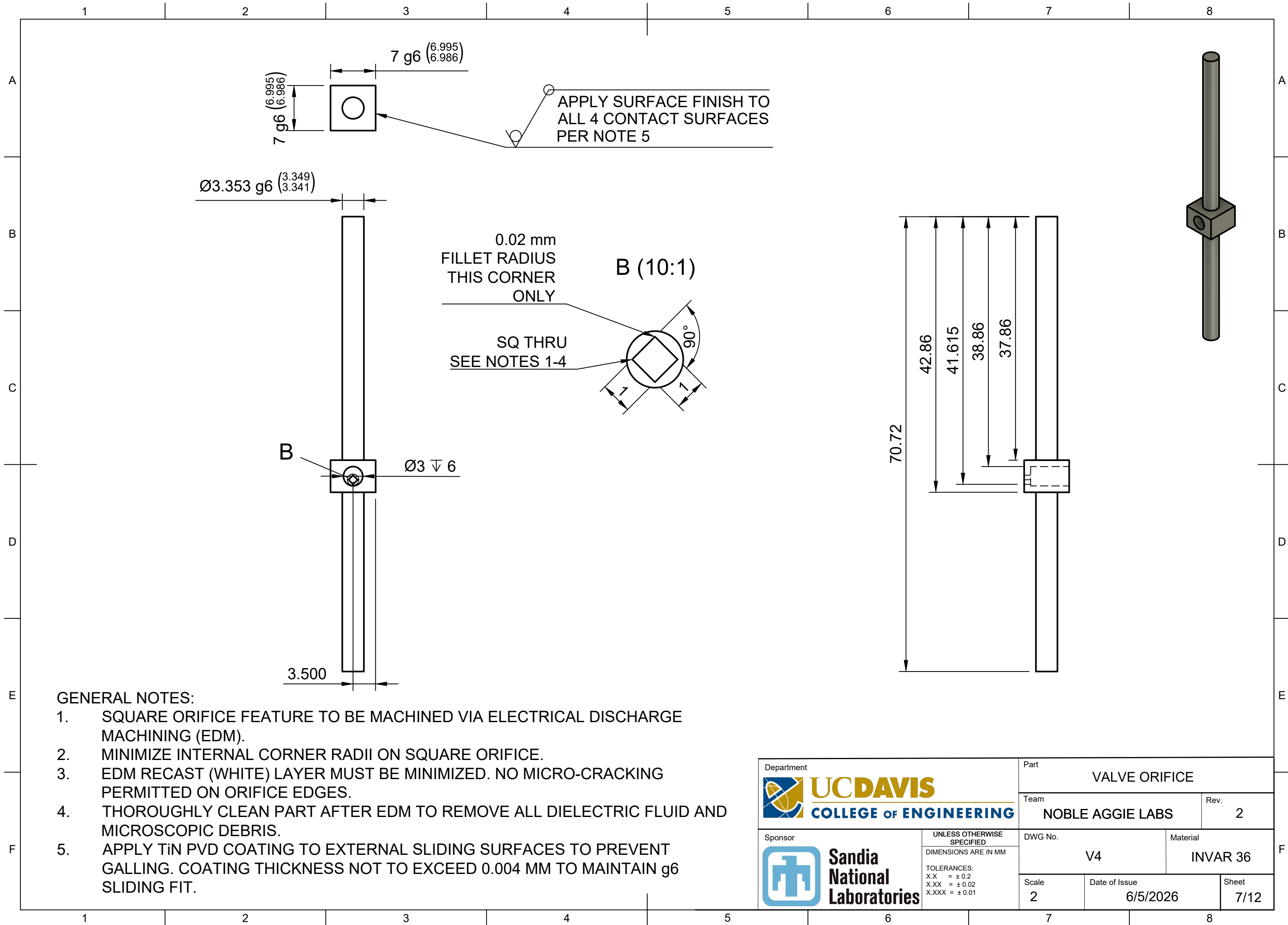
Department		Part		
		VALVE SUB-ASSEMBLY		
Team		NOBLE AGGIE LABS		Rev. 2
Sponsor		DWG No.	Material	
		V0		
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM		Scale	Date of Issue	Sheet
TOLERANCES: X.X = ± 0.2 X.XX = ± 0.02 X.XXX = ± 0.01		1	6/5/2026	3/12

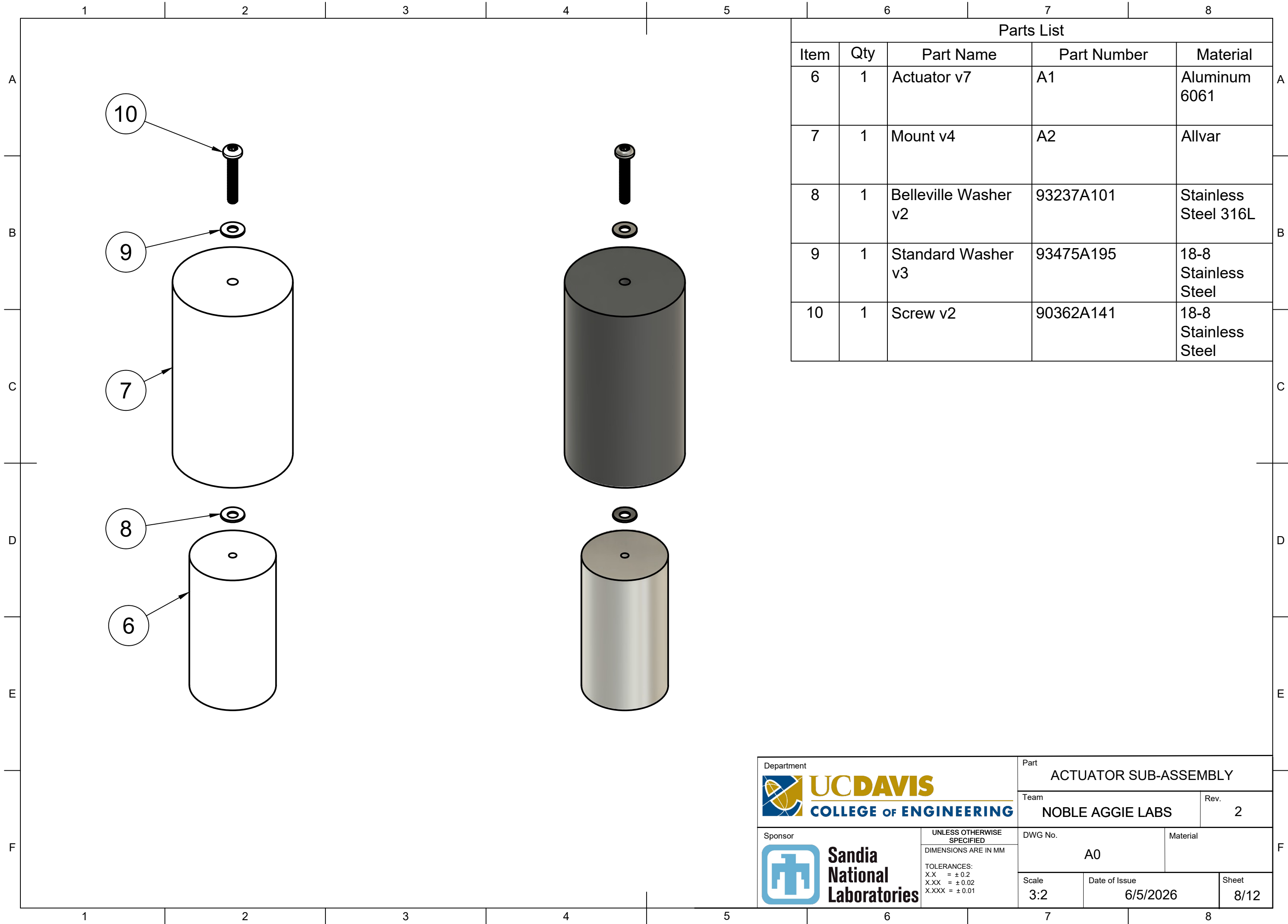


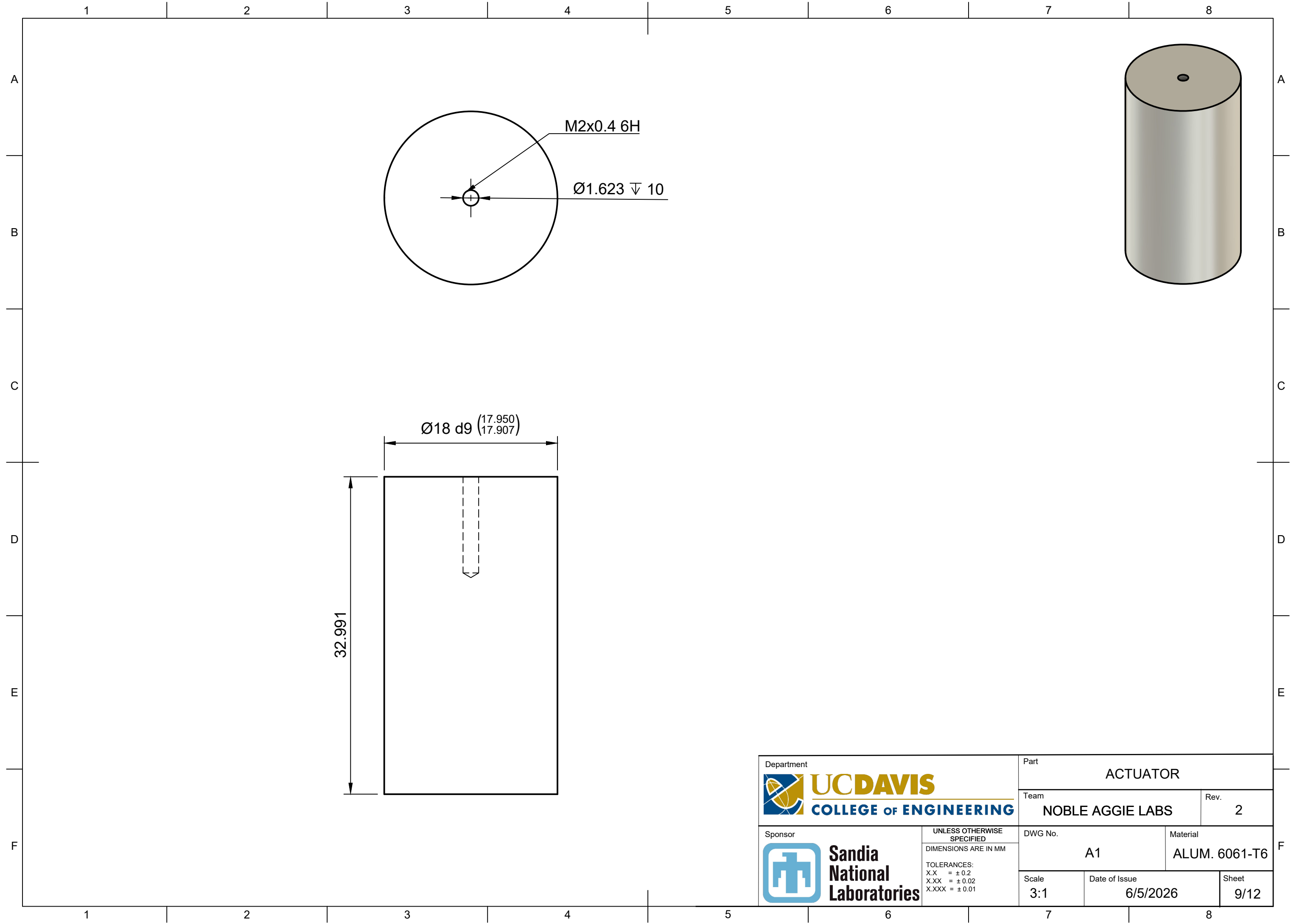
Department		Part		
		HOUSING CENTER		
		Team	NOBLE AGGIE LABS	Rev. 2
Sponsor		DWG No.		Material
		V1		INVAR 36
		Scale	Date of Issue	Sheet
		3:1	6/5/2026	4/12

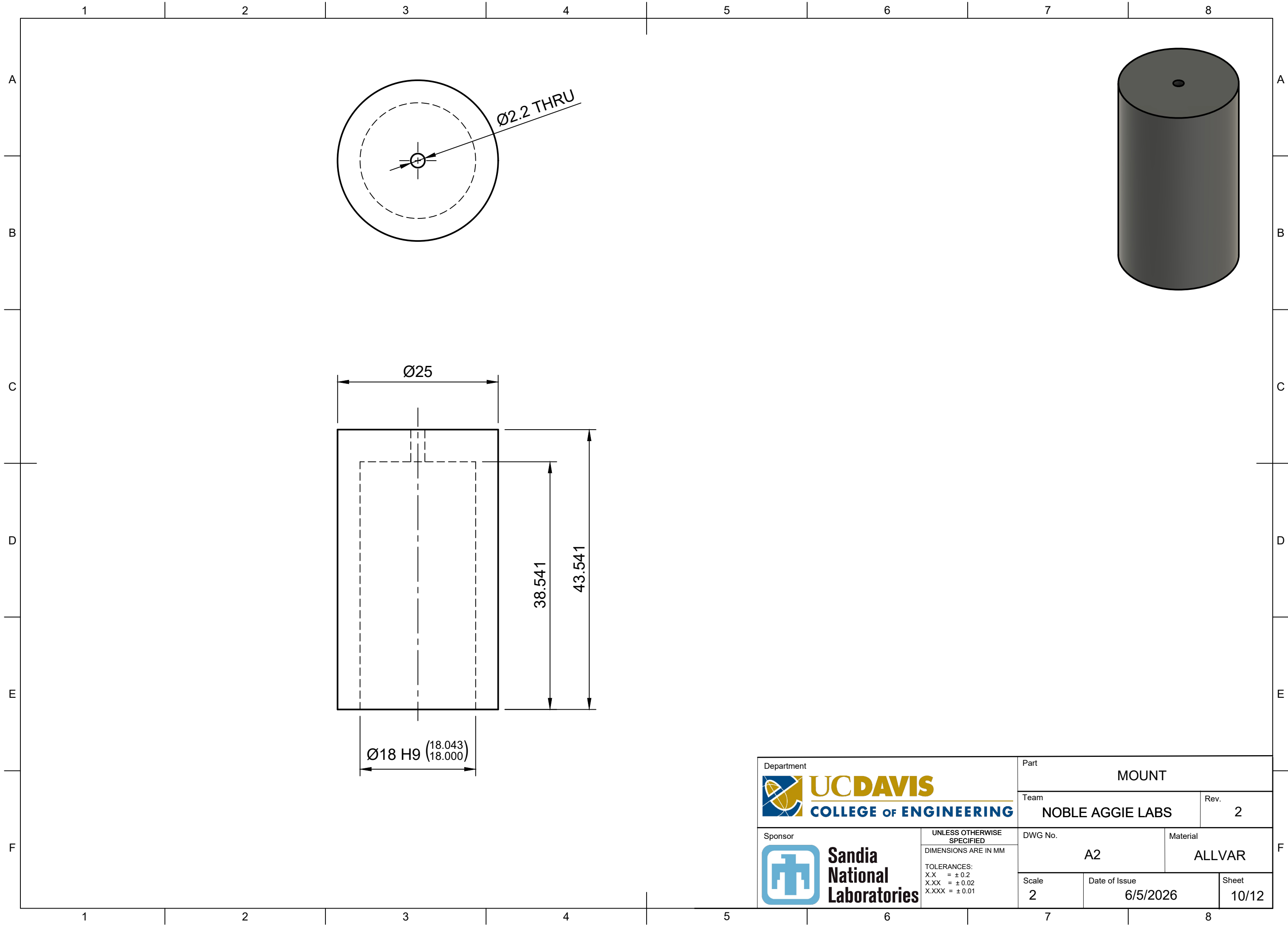








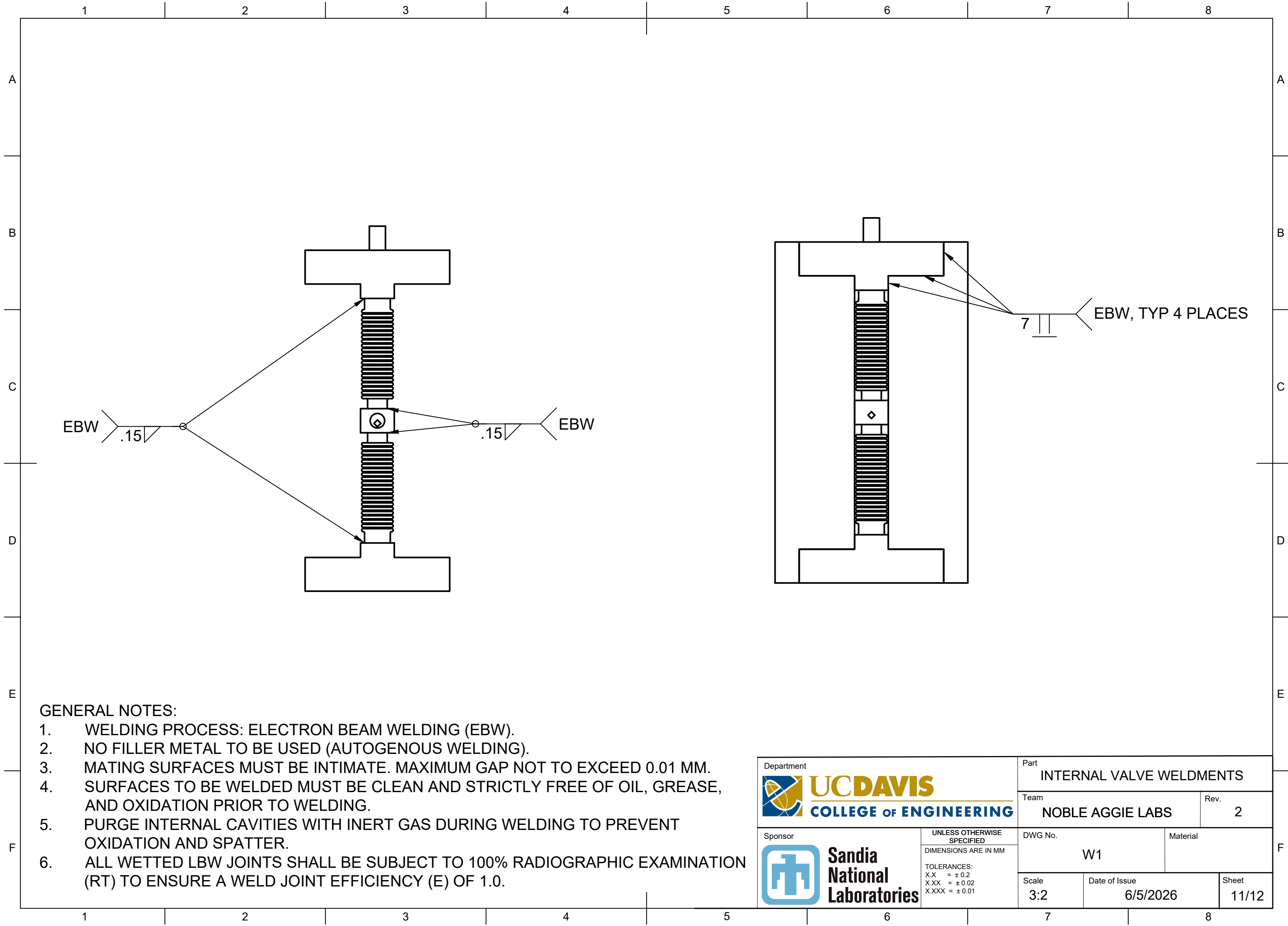




Department		Part		
		MOUNT		
		Team	NOBLE AGGIE LABS	Rev. 2
Sponsor		DWG No.		Material
		A2		ALLVAR
		Scale	Date of Issue	Sheet
		2	6/5/2026	10/12

UNLESS OTHERWISE
SPECIFIED
DIMENSIONS ARE IN MM

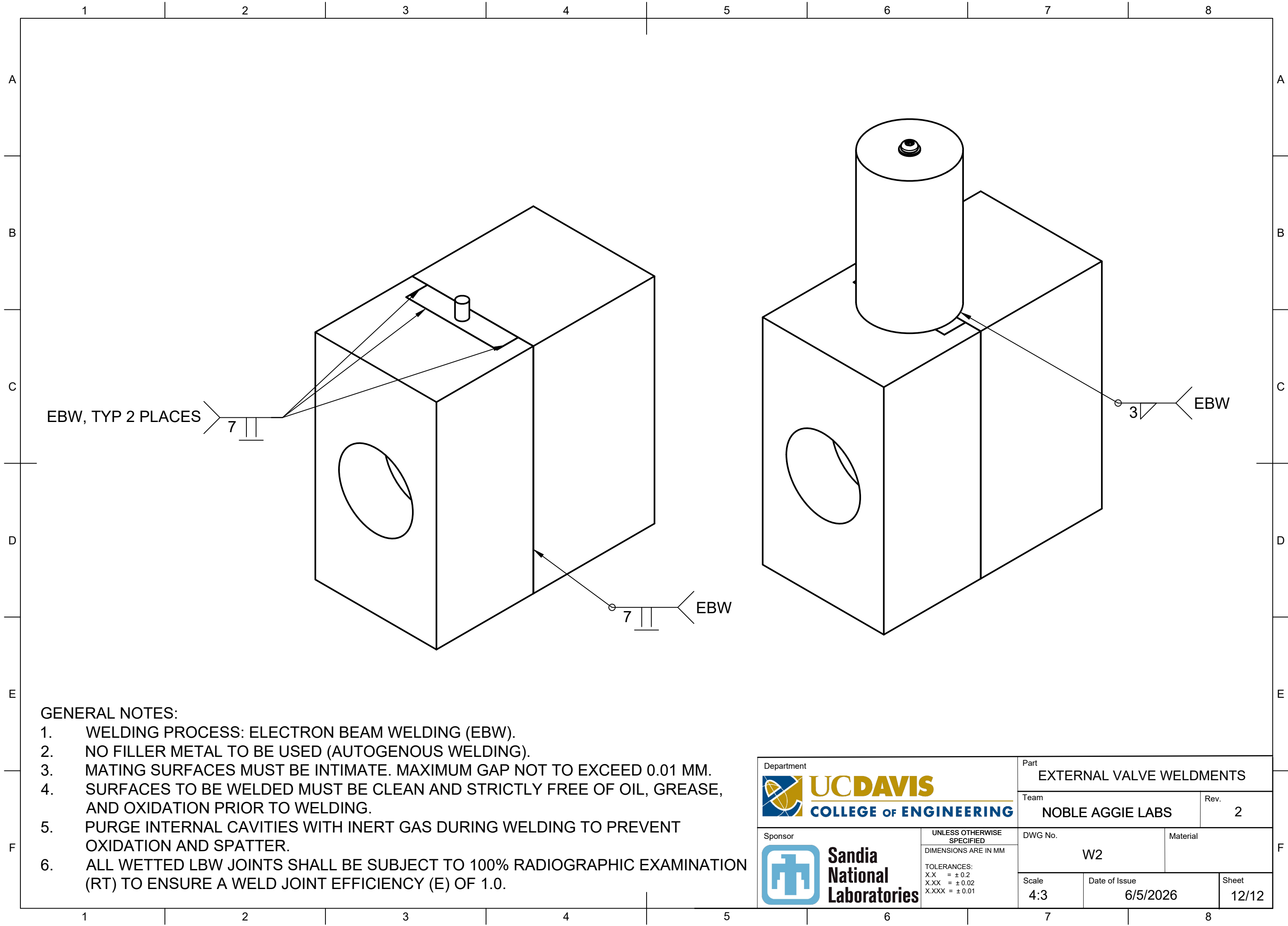
TOLERANCES:
X.X = ± 0.2
X.XX = ± 0.02
X.XXX = ± 0.01



GENERAL NOTES:

1. WELDING PROCESS: ELECTRON BEAM WELDING (EBW).
2. NO FILLER METAL TO BE USED (AUTOGENOUS WELDING).
3. MATING SURFACES MUST BE INTIMATE. MAXIMUM GAP NOT TO EXCEED 0.01 MM.
4. SURFACES TO BE WELDED MUST BE CLEAN AND STRICTLY FREE OF OIL, GREASE, AND OXIDATION PRIOR TO WELDING.
5. PURGE INTERNAL CAVITIES WITH INERT GAS DURING WELDING TO PREVENT OXIDATION AND SPATTER.
6. ALL WETTED LBW JOINTS SHALL BE SUBJECT TO 100% RADIOGRAPHIC EXAMINATION (RT) TO ENSURE A WELD JOINT EFFICIENCY (E) OF 1.0.

Department		Part		
		INTERNAL VALVE WELDMENTS		
Sponsor		Team		Rev.
		NOBLE AGGIE LABS		2
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM		DWG No.	Material	
TOLERANCES: X.X = ± 0.2 X.XX = ± 0.02 X.XXX = ± 0.01		W1		
		Scale	Date of Issue	Sheet
		3:2	6/5/2026	11/12



GENERAL NOTES:

1. WELDING PROCESS: ELECTRON BEAM WELDING (EBW).
2. NO FILLER METAL TO BE USED (AUTOGENOUS WELDING).
3. MATING SURFACES MUST BE INTIMATE. MAXIMUM GAP NOT TO EXCEED 0.01 MM.
4. SURFACES TO BE WELDED MUST BE CLEAN AND STRICTLY FREE OF OIL, GREASE, AND OXIDATION PRIOR TO WELDING.
5. PURGE INTERNAL CAVITIES WITH INERT GAS DURING WELDING TO PREVENT OXIDATION AND SPATTER.
6. ALL WETTED LBW JOINTS SHALL BE SUBJECT TO 100% RADIOGRAPHIC EXAMINATION (RT) TO ENSURE A WELD JOINT EFFICIENCY (E) OF 1.0.

Department		Part	
		EXTERNAL VALVE WELDMENTS	
Team		Rev.	
NOBLE AGGIE LABS		2	
Sponsor		DWG No.	
		W2	
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM		Material	
TOLERANCES: X.X = ± 0.2 X.XX = ± 0.02 X.XXX = ± 0.01		Scale	
		4:3	
		Date of Issue	
		6/5/2026	
		Sheet	
		12/12	